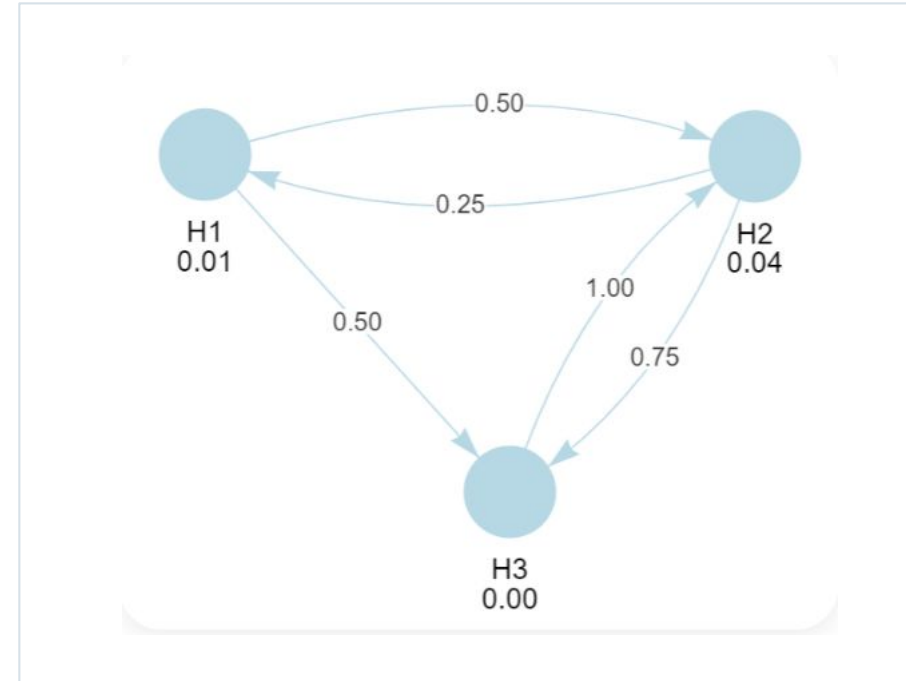


SDSS 2026

Interactive Visualization of Graphical Multiple Testing Procedures

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An R Shiny Interface for Clinical Trial Design

Why This Problem Matters

Confirmatory trials

- Several efficacy questions are often tested together.
- Typical oncology designs combine endpoints and populations.
- One false claim is enough to compromise the study conclusion.
- Regulators therefore expect strong FWER control.



Main Gap

Graphical multiple testing procedures are statistically powerful, but logic can be difficult to review and execute without software support

Presentation Goals

- Show how graphMTP translates graphical multiple testing procedures into an interactive R Shiny workflow

→ We will cover:

1. The statistical problem: FWER and multiplicity
2. The theory: closed testing, consonance, and graphical procedures
3. The app: alpha flow, boundary review, and analysis history
4. Three examples showing app-native designs

Multiplicity Inflates Type I Error

- **Testing many hypotheses increases the chance of at least one false rejection.**
 - For m independent tests at level α :
 - $P(\geq 1 \text{ false rejection}) = 1 - (1 - \alpha)^m$
 - **Examples:**
 - $m = 5, \alpha = 0.05$: FWER ≈ 0.226
 - $m = 20, \alpha = 0.05$: FWER ≈ 0.64
- Controlling each test marginally at 0.05 does not control the family-level error rate.

Sources of Multiplicity in Clinical Trials

Across endpoints

- PFS: progression-free survival
- OS: overall survival
- Primary and key secondary endpoints

Across populations

- Biomarker-positive
- Overall intent-to-treat (ITT)
- Subgroups and enrichment designs

Across time

- Interim analyses
- Final analyses
- Repeated testing of the same hypothesis

Clinical Hierarchy Changes the Design Target

What the clinical team wants

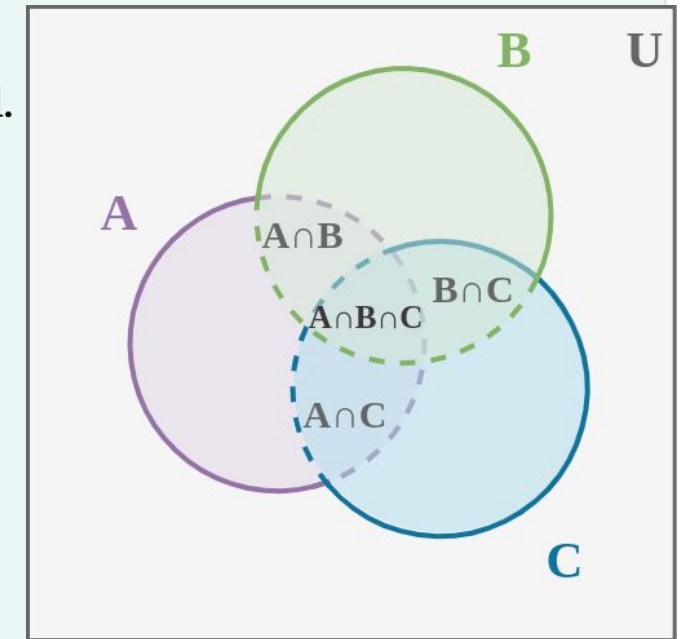
- Primary claims should be protected first.
- Secondary claims should open only after meaningful success.
- Different branches assigned different initial priority
- A procedure that reflects the medical story

What the statistician needs

- Strong FWER control
- Pre-specified testing rules
- A procedure that is reproducible at analysis time.
- A clear path from design logic to software implementation.

Closed Testing Principle

- **Closed testing provides the theoretical foundation for strong FWER control.**
- For elementary hypotheses $H_1, \dots, H_m$, define intersection hypotheses:
 - $H_J = \bigcap_{j \in J} H_j$, where $J \subseteq \{1, \dots, m\}$.
- **Rejection rule:**
 - Reject H_i only if every intersection hypothesis containing H_i is rejected.
 - This gives strong FWER control at level α .
- **Operational challenge:** $2^m - 1$
- intersection hypotheses must be considered.
- Examples:
 - $m = 4$: 15 intersections
 - $m = 8$: 255 intersections



Consonance and the Stepwise Shortcut

- **Consonance makes closed testing operational.**
 - A procedure is consonant if rejecting an intersection hypothesis implies rejecting at least one elementary hypothesis inside that intersection.
 - **Why this matters:**
 - Without consonance, closed testing may require explicit intersection testing.
 - With consonance, the procedure can be run stepwise.
 - This creates the logic behind sequentially rejective graphical procedures.
- Graphical procedures are designed so that the shortcut remains valid while avoiding explicit closure-table enumeration.

Formal Graph Object

Mathematical object $G = (\alpha, G)$

- $\alpha = (\alpha_1, \dots, \alpha_m)$: local significance levels
- $\sum_i \alpha_i \leq \alpha$
- $G = [g_{ij}]$: transition matrix
- g_{ij} : fraction of α_i transferred to H_j if H_i rejects
- $g_{ii} = 0$: no self-loops
- $\sum_j g_{ij} \leq 1$: outgoing weights cannot exceed one

App object

- Nodes table: stores current α_i for each hypothesis.
- Edges table: stores each nonzero g_{ij} entry.
- Graph : makes α and G visually reviewable.
- JSON import/export: preserves full (V, α, G) state for replay, handoff, and audit.
- Both tables update live as hypotheses are rejected.

Algorithm 1: Graph Update Procedure (Bretz et al. 2009)

Input: $G = (\alpha, G)$ with $\alpha_i \geq 0$, $\sum \alpha_i \leq \alpha$, $g_{ii} = 0$, $\sum_{j \neq i} g_{ij} \leq 1$.

Step 1 — Test:

For each active H_i : if $p_i \leq \alpha_i$, reject H_i (go to Step 2).

If no rejection possible, stop.

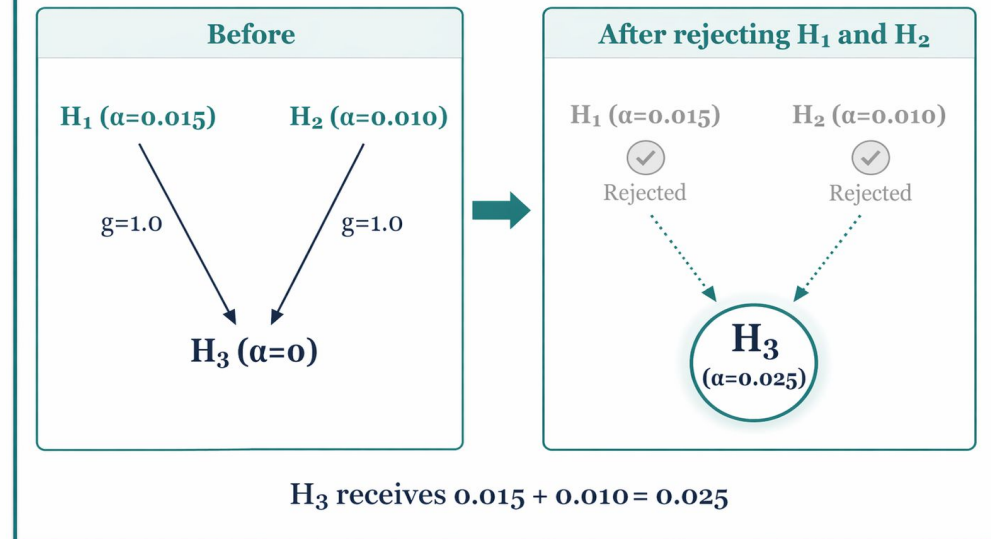
Step 2 — Recycle: for all $j \neq i$,

$$\alpha_j := \alpha_j + \alpha_i \cdot g_{ij}$$

$$g_{kj} := (g_{kj} + g_{ki} \cdot g_{ij}) / (1 - g_{ki} \cdot g_{ij}) \text{ for } k \neq i$$

Step 3 — Remove: delete H_i from graph. Return to Step 1.

Complementary Example



Group Sequential Extension

- Each hypothesis H_i may be tested at multiple planned analysis times.
- For hypothesis H_i , a spending function $\alpha_i(t)$ allocates local alpha over information fraction $t \in [0, 1]$.
- At look k , the incremental alpha is:
- $\delta_{ik} = \alpha_i(t_{ik}) - \alpha_i(t_{i,k-1})$
- **Decision rule:**
 - Test H_i at the planned look using the available alpha increment.
 - If H_i rejects, remaining alpha can be recycled to active hypotheses.
 - The graph handles multiplicity across hypotheses.
 - Spending functions handle multiplicity across time.

Spending Functions and Repeated P-Values

- **Lan Demets O'Brien-Fleming (OF):**

$$\alpha_i(t) \approx 2[1 - \Phi(z_{\{\alpha/2\}/\sqrt{t}})].$$

- Spends very little early; conservative at interim looks.

- **Pocock:** $\alpha_i(t) \approx \alpha \cdot \log(1 + (e - 1)t)$.

- Spends more uniformly

- **Hwang-Shih-DeCani(γ):**

$$\alpha_i(t) = \alpha(1 - e^{\{-\gamma t\}})/(1 - e^{\{-\gamma\}}).$$

Interpolates between OF ($\gamma \rightarrow -\infty$) and Pocock ($\gamma \rightarrow 0$).

→ Spending rule is hypothesis-specific in the group sequential app.

Paper-to-App Translation Layer

Paper → App translation

- Local level $\alpha_i \rightarrow$ node alpha value in the graph.
- Transition weight $g_i \square \rightarrow$ directed edge.
- Spending function $\alpha_i(t) \rightarrow$ boundary preview table.
- Repeated look decisions $\rightarrow$ frozen analysis-history rows.
- Intersection weight $w_i(J) \rightarrow$ computed live by the rejection engine.
- Rejected set at each step $\rightarrow$ reded-out nodes.

CASE A

Maurer and Bretz Four-Hypothesis Example

Goal: show how a clinically structured graphical procedure translates directly into the app.

Graphical Multiple Testing

Home Design Group Sequential Design Analysis

Graph Design

[Hide data panel](#)

Create Object Hypothesis H1 Reject Selected Clear Results Auto Layout

Use these controls for classic graphical testing. Open Group Sequential Design to plan interim analyses, and Analysis to submit one-sided results by analysis time.

Nodes

hypothesis	alpha
H1	0.0125
H2	0.0125
H3	0
H4	0

Edges

from	to	weight
H1	H2	0.5
H1	H3	0.5
H2	H1	0.5
H2	H4	0.5
H3	H4	1
H4	H3	1

Import JSON graph

Browse... case_a_maurer_bretz_2c Upload complete

Export graph

Build the graph here. Group Sequential Design adds the study-design tables and boundary schedule. Analysis submits one-sided results one analysis time at a time.

Output

Case A Hypotheses and Clinical Hierarchy

Hypotheses

- H1: PFS in the biomarker-positive population.
- H2: OS in the biomarker-positive population.
- H3: PFS in the ITT population.
- H4: OS in the ITT population.

Hierarchy

- Biomarker-positive claims receive the initial alpha.
- ITT claims open only after success in the biomarker-positive branch.
- Within a population, PFS is placed ahead of OS.
- This creates both cross-population and within-population gatekeeping.

Case A: Initial Graph $G = (\alpha, G)$

- Total one-sided FWER level $\alpha = 0.025$.
- Initial local levels: $\alpha = (0.0125, 0.0125, 0, 0)$.

Biomarker-positive hypotheses carry the full budget; ITT starts at zero.

- Transition matrix G (nonzero entries, row $\times$ col):

$$g_{12} = 0.5 \text{ (H1} \rightarrow \text{H2)}, \quad g_{13} = 0.5 \text{ (H1} \rightarrow \text{H3)}$$

$$g_{21} = 0.5 \text{ (H2} \rightarrow \text{H1)}, \quad g_{24} = 0.5 \text{ (H2} \rightarrow \text{H4)}$$

$$g_{34} = 1.0 \text{ (H3} \rightarrow \text{H4)}, \quad g_{43} = 1.0 \text{ (H4} \rightarrow \text{H3)}$$

- Row sums ≤ 1 ; diagonal = 0 ; budget conservation holds.

Case A Click-Through: Build the Nodes

My Application

Home

Design

Directed Graph Designer

[Hide data panel](#)

Create Test ObjectReject Selected HypothesisEdit Mode

Select hypothesis to reject

Nodes

hypothesis	alpha
H1	0
H2	0
H3	0

Edges

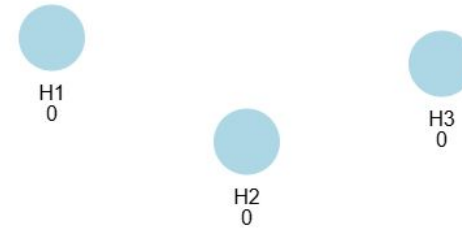
from	to	weight
No data available in table		

Import JSON graph

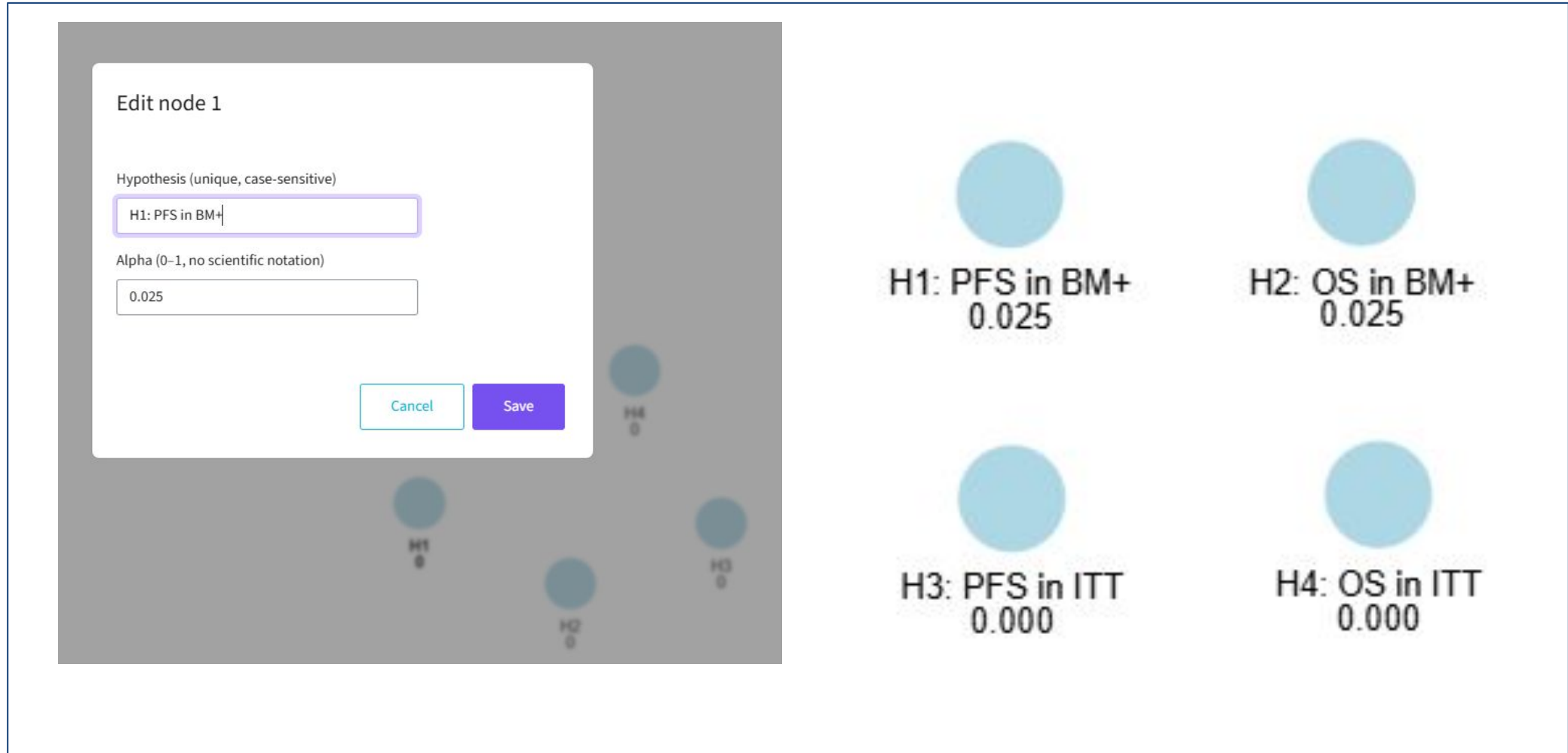
Browse...

No file selected

Export graph



Case A Click-Through: Build the Nodes



The image shows a software interface for building nodes. On the left, a dialog box titled "Edit node 1" is open. It contains two input fields: "Hypothesis (unique, case-sensitive)" with the text "H1: PFS in BM+" and "Alpha (0-1, no scientific notation)" with the value "0.025". Below these fields are "Cancel" and "Save" buttons. In the background, a diagram shows four nodes, each represented by a blue circle. The nodes are labeled as follows:

- Top-left: H1: PFS in BM+ 0.025
- Top-right: H2: OS in BM+ 0.025
- Bottom-left: H3: PFS in ITT 0.000
- Bottom-right: H4: OS in ITT 0.000

Case A Click-Through: Build the Edges

The screenshot displays a graph building interface. On the left, four nodes are arranged in a 2x2 grid:

- Top-left node: H1: PFS 0.0
- Top-right node: BM+ 5
- Bottom-left node: H3: PFS in ITT 0.000
- Bottom-right node: H4: OS in ITT 0.000

A context menu is open over the top-left node (H1), showing two options:

- Delete this node
- Start edge from here

On the right, a dialog box titled "New edge: 1 → 2" is shown. It contains a label "Weight (0-1, no scientific notation)" and a text input field with the value "0.5". At the bottom right of the dialog are two buttons: "Cancel" and "Save".

Case A Click-Through: Import the Graph



**If you already have a graph saved,
you can upload it**



**Export your graph once you are done
with designing**

Build the graph here. Group Sequential Design adds the study-design tables and boundary schedule. Analysis submits one-sided results one analysis time at a time.

Output

Case A Click-Through: Import the Graph

Graphical Multiple Testing [Home](#) [Design](#) [Group Sequential Design](#) [Analysis](#)

Graph Design

[Hide data panel](#)

Hypothesis

Create Object H1 Reject Selected Clear Results Auto Layout

Use these controls for classic graphical testing. Open Group Sequential Design to plan interim analyses, and Analysis to submit one-sided results by analysis time.

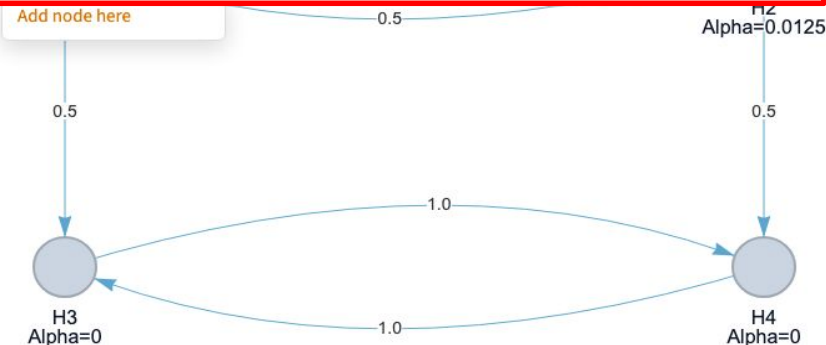
Nodes

hypothesis	alpha
H1	0.0125
H2	0.0125
H3	0
H4	0

Edges

from	to	weight
H1	H2	0.5
H1	H3	0.5
H2	H1	0.5
H2	H4	0.5
H3	H4	1
H4	H3	1

Nodes and Edges auto-populated as you make the graph



Case A Click-Through: Create Object

Graphical Multiple Testing [Home](#) [Design](#) [Group Sequential Design](#) [Analysis](#)

Graph Design

[Hide data panel](#)

Create Object

Hypothesis

H1

Reject Selected

Use these controls for classic graphical testing. Open Group Sequential Design to plan int

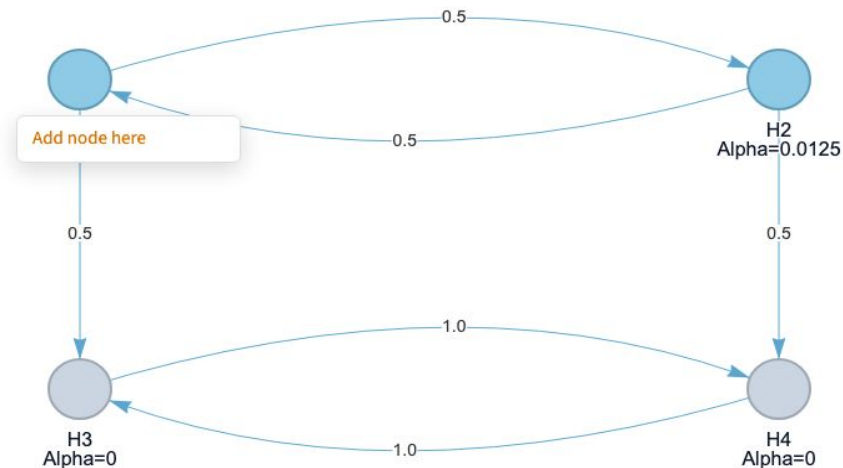
Once you have finalized your design,
click “Create Object”

Nodes

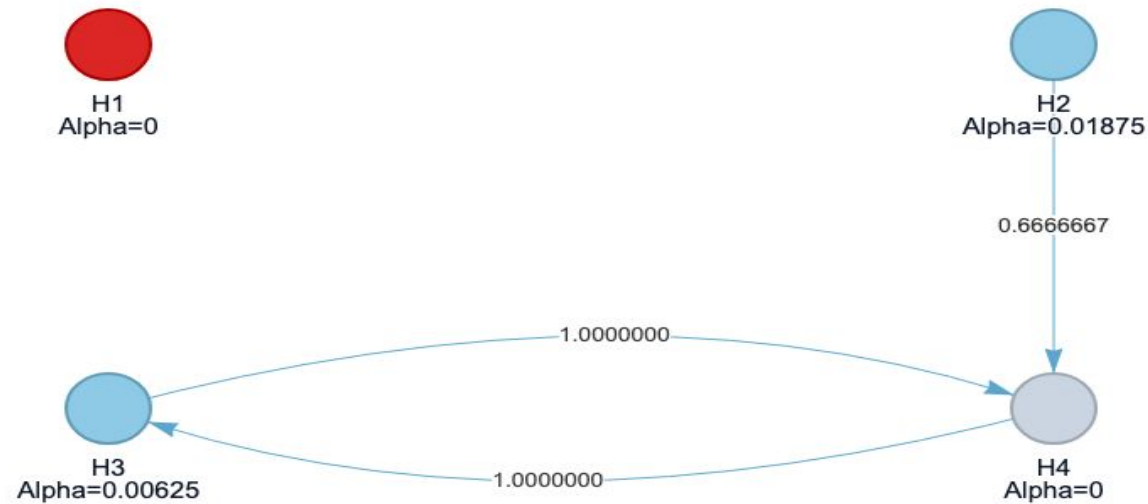
hypothesis	alpha
H1	0.0125
H2	0.0125
H3	0
H4	0

Edges

from	to	weight
H1	H2	0.5
H1	H3	0.5
H2	H1	0.5
H2	H4	0.5
H3	H4	1
H4	H3	1



Case A Step 1 in the App



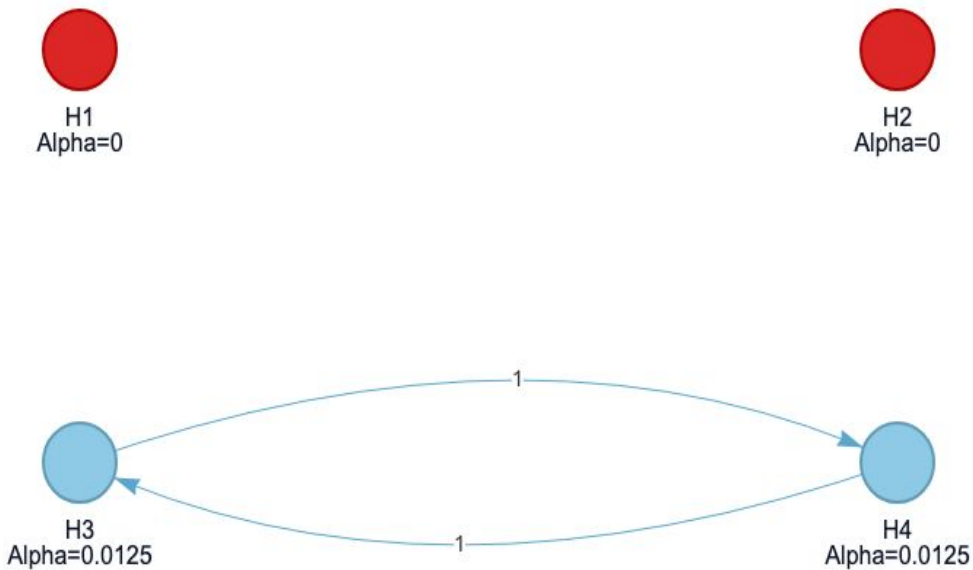
Case A Step 2 in the App

Nodes

hypothesis	alpha
H1	0.0125
H2	0.0125
H3	0
H4	0

Edges

from	to	weight
H1	H2	0.5
H1	H3	0.5
H2	H1	0.5
H2	H4	0.5
H3	H4	1
H4	H3	1



Case A Step 3 and Final State

Graph Design

[Hide data panel](#)

Create Object

Hypothesis
H4

Reject Selected

Clear Results

Auto Layout

Use these controls for classic graphical testing. Open Group Sequential Design to plan interim analyses, and Analysis to submit one-sided results by analysis time.

Nodes

hypothesis	alpha
H1	0.0125
H2	0.0125
H3	0
H4	0

Edges

from	to	weight
H1	H2	0.5
H1	H3	0.5
H2	H1	0.5
H2	H4	0.5
H3	H4	1
H4	H3	1

Import JSON graph

Browse...

case_a_maurer_bretz_20

Upload complete



H1
Alpha=0



H2
Alpha=0



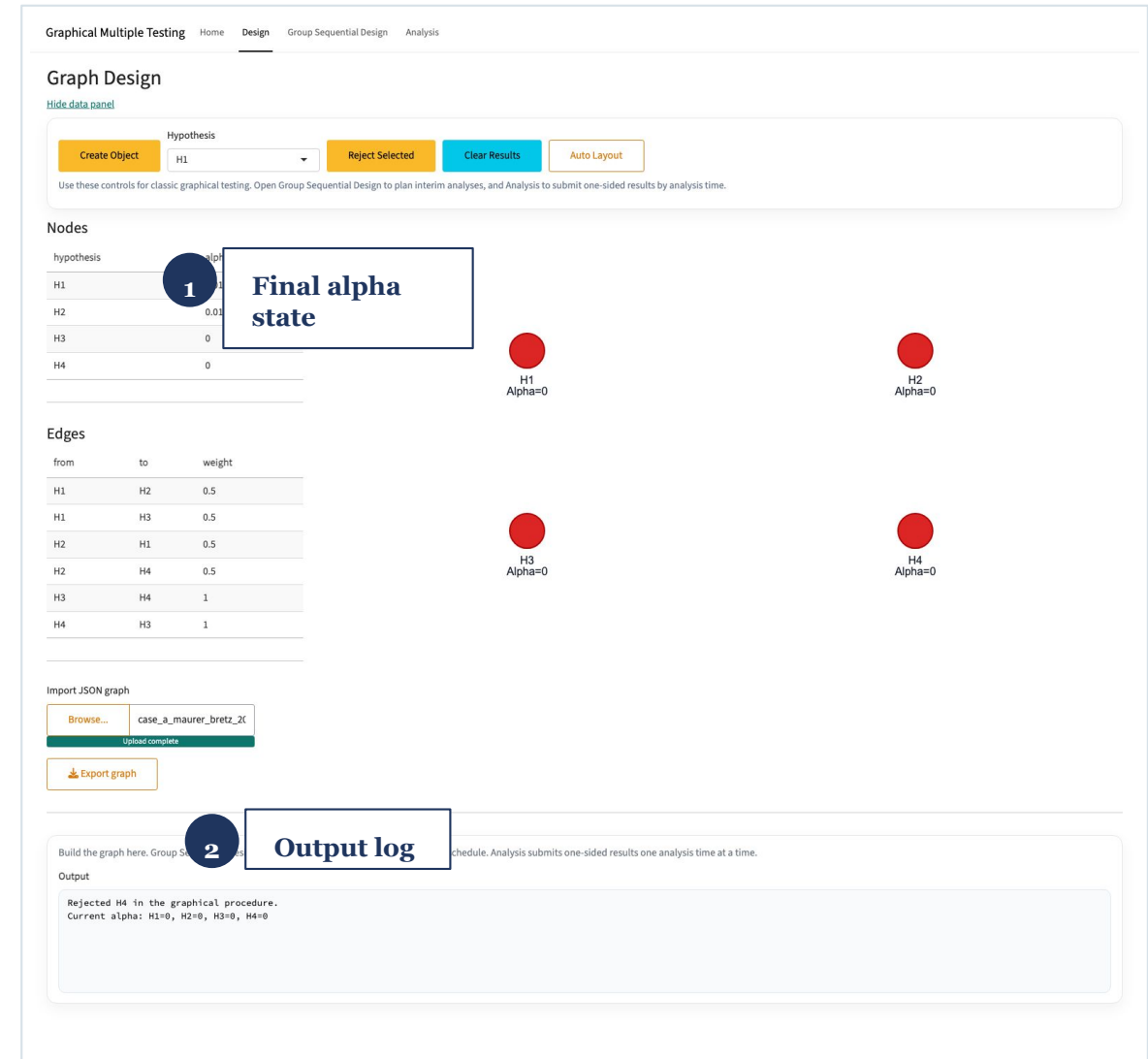
H3
Alpha=0



H4
Alpha=0.025

Case A Final State in the App

- H4 is now tested with the full remaining alpha.
- The final rejection closes the active graph.
- The app displays both the final node state and the cumulative output log.



CASE B

One Interim Analysis with Same-Round Recycling

Goal: show how planned looks, boundaries, and frozen analysis history interact.

Graphical Multiple Testing

Home

Design

Group Sequential Design

Analysis

Locked design

Hypotheses: H1, H2, H3. Analysis times: 2. Design signature applied.

Analysis

Submit one full analysis time at a time. Enter one-sided p-values for every hypothesis scheduled at that analysis time and the app will update the graph and alpha recycling together.

Analysis Time = shared trial checkpoint. Interim Analysis (Look) = hypothesis-specific test at that checkpoint.

Reset Analysis State

Open live graph and activity

Submit One Analysis Time

Choose an analysis time, then enter one-sided p-values for every active hypothesis scheduled at that analysis time.

Analysis Time

Analysis Time 2

Hypothesis	Interim Analysis (Look)	Current Alpha	Planned Information Fraction	Planned Max Info	Observed Info	Previous Saved Result	Boundary p	Entered One-Sided p	Decision
H3	Final Analysis 1 of 1	0.025	1	100	100	No prior saved result	0.025		Computed on submit

Submit Analysis Time

Saved analysis time 2 with 2 saved results across 2 hypotheses.

Live Analysis State

Use this table to track each hypothesis, the latest submitted result, and the next scheduled analysis time.

Hypothesis	Current Alpha	Last Submitted Analysis Time / Look	Last Observed p	Decision	In Graph	Testable	Next Analysis Time
H1	0	Analysis Time 2 / Look 2	0.0001	Reject	No	No	—
H2	0	Analysis Time 2 / Look 2	0.02	Reject	No	No	—
H3	0.025			Pending	Yes	Yes	2

Submitted Analyses

Submission is the saved batch. Analysis Time / Look identifies the specific checkpoint and hypothesis-specific look inside that batch. One hypothesis can appear more than once in the same submission when same-analysis alpha recycling triggers a retest. Alpha At Submission is frozen for each saved event row and does not change after recycling.

Submission	Analysis Time / Look	Hypothesis	Alpha Spending	Info Fraction	Alpha At Submission	P	Boundary p	Boundary z	Decision
Submission 1	Analysis Time 1 / Look 1	H1	Lan-DeMets O'Brien-Fleming	0.5	0.01	0.01	0.000269717	3.46037	Do not reject
Submission 1	Analysis Time 1 / Look 1	H2	Lan-DeMets O'Brien-Fleming	0.5	0.015	0.01	0.0005819217	3.247596	Do not reject
Submission 2	Analysis Time 2 / Look 2 (final)	H1	Lan-DeMets O'Brien-Fleming	1	0.01	0.0001	0.0099070116	2.329851	Reject
Submission 2	Analysis Time 2 / Look 2 (final)	H2	Lan-DeMets O'Brien-Fleming	1	0.0225	0.02	0.0223499396	2.007468	Reject

Current app status for this section: workflow is available for demonstration and review; validation remains in progress.

Case B Design Object and Planned Looks

Design

- **Initial local alphas**
 - $H_1 = 0.010$
 - $H_2 = 0.015$
 - $H_3 = 0$
- **Transition weights**
 - $H_1 \rightarrow H_2: 0.75$
 - $H_1 \rightarrow H_3: 0.25$
 - $H_2 \rightarrow H_1: 0.50$
 - $H_2 \rightarrow H_3: 0.50$

Sequential plan

- **Planned looks**
 - H_1 : interim and final
 - H_2 : interim and final
 - H_3 : final only

Spending rule:

Lan-Demets O'Brien-Fleming

Case B Click-Through: Hypothesis Plan

Graphical Multiple Testing

Home

Design

Group Sequential Design

Analysis

Group Sequential Design

Walk through each step to define the group sequential design. You can go back to any previous step before finalizing.

Current graph hypotheses: H1, H2, H3. This design currently spans 2 analysis times and us

1 Hypothesis Plan

2 Analysis Timing

3 Boundary Review

Step 1. Hypotheses and Alpha Spending Functions

Each row comes from the Design graph. Set the planned looks (K) for that hypothesis, choose the alpha spending function, and enter parameters when needed.

☐ Override planned maximum information

Hypothesis	Planned Analyses (K)	Alpha Spending Function	Rule Parameters
H1	2	Lan-DeMets O'Brien-Fleming	Auto from selected rule
H2	2	Lan-DeMets O'Brien-Fleming	Auto from selected rule
H3	1	Lan-DeMets O'Brien-Fleming	

Planned maximum information stays at the internal default unless you override it.

Reset Defaults

Next: Analysis Timing →

1. Enter the number of interim analysis you are planning to do

2. Choose the alpha spending function

Case B Click-Through: Hypothesis Plan

Graphical Multiple Testing

Home

Design

Group Sequential Design

Analysis

Group Sequential Design

Walk through each step to define the group sequential design. You can go back to any previous step before finalizing.

Current graph hypotheses: H1, H2, H3. This design currently spans 2 analysis times and uses one-sided testing throughout.

1 Hypothesis Plan

2 Analysis Timing

3 Boundary Review

Step 1. Hypotheses and Alpha Spending Functions

Each row comes from the Design graph. Set the planned looks (K) for that hypothesis, choose the alpha spending function, and enter parameters when needed.

☐ Override planned looks

Hypothesis	Planned looks (K)	Alpha spending function
H1		
H2	2	Lan-DeMets O'Brien-Fleming
H3	1	Lan-DeMets O'Brien-Fleming

Planned maximum information stays at the internal default unless you override it.

Reset Defaults

Next: Analysis Timing →

3. If you need to reset, click this. It would automatically click reset number of analyses to 1

4. Once you are done with your hypothesis and alpha spending function, click this

Case B Click-Through: Information Schedule

Group Sequential Design

Walk through each step to define the group sequential design. You can click to go directly to any step.

Current graph hypotheses: H1, H2, H3. This



Hypothesis Plan

3

Boundary Review

1. Edit the analysis time per hypothesis and interim looks. This can be any reasonable number

Step 2. Planned Analysis Times for Entire Study

The schedule is generated from the hypothesis plan. Adjust the analysis time or information fraction when a hypothesis needs a different timing.

Analysis Time	Hypothesis	Interim Analysis (Look)	Information Fraction
1	H1	Interim Analysis 1 of 2	0.5
1	H2	Interim Analysis 1 of 2	0.5
2	H1	Final Analysis 2 of 2	1
2	H2	Final Analysis 2 of 2	1
2	H3	Final Analysis 1 of 1	1

2. Make sure to check corresponding IF after you change the analysis time

← Back: Hypotheses and Alpha Spending Functions

Next: Boundary Review →

Case B Click-Through: Boundary Preview

Group Sequential Design

Walk through each step to define the group sequential design. You can go back to any previous step before finalizing.

Current graph hypotheses: H1, H2, H3. This design currently spans 2 analysis times and uses one-sided testing throughout.

✓ Hypothesis Plan

✓ Analysis Timing

3 Boundary Review

Step 3. Boundary Review

Review the design-time one-sided boundaries. When everything looks correct, finalize the design to proceed to the Analysis tab.

This table is a design-time preview only. When you later submit one Analysis Time, TrialSimulator resolves the live batch, including any same-analysis alpha recycling, and the Submitted Analyses table freezes the package-truth boundary used for each saved result.

Analysis Time	Hypothesis	Interim Analysis (Look)	Total Analyses	Info Fraction	Alpha Spending	Current Alpha	Look Alpha	Cumulative Alpha	Boundary p	Boundary z	Status
1	H1	1	2	0.5	Lan-DeMets O'Brien-Fleming	0.01	0.000269717	0.000269717	0.000269717	3.46037	Ready
1	H2	1	2	0.5	Lan-DeMets O'Brien-Fleming	0.015	0.0005819217	0.0005819217	0.0005819217	3.247596	Ready
2	H1	2	2	1	Lan-DeMets O'Brien-Fleming	0.01	0.0097302911	0.010000008	0.0099075755	2.32983	Ready
2	H2	2	2	1	Lan-DeMets O'Brien-Fleming	0.015	0.0144180849	0.0150000067	0.014804171	2.175291	Ready
2	H3	1	1	1	Lan-DeMets O'Brien-Fleming	0					Not currently testable

← Back: Analysis Timing

Finalize Design

Case B Click-Through: Enter P-Values

[Graphical Multiple Testing](#) [Home](#) [Design](#) [Group Sequential Design](#) [Analysis](#)

Locked design
Hypotheses: H1, H2, H3. Analysis times: 2. Design signature applied.
[Edit Design](#)

Analysis

Submit one full analysis time at a time. Enter one-sided p-values for every hypothesis scheduled at that analysis time and the app will update the results.

Analysis Time = shared trial checkpoint. Interim Analysis (Look) = hypothesis-specific test at that checkpoint.

[Open live graph and activity](#)

Submit One Analysis Time

Choose an analysis time, then enter one-sided p-values for every active hypothesis scheduled at that analysis time.

Analysis Time

Analysis Time 1

Hypothesis	Interim Analysis (Look)	Current Alpha	Planned Information Fraction	Planned Max Info	Observed Info	Previous Saved Result	Boundary p	Enter 1-Sided p	Decision
H1	Interim Analysis 1 of 2	0.01	0.5	100	50	No prior saved result	0.000269717	0.01	Computed on submit
H2	Interim Analysis 1 of 2	0.015	0.5	100	50	No prior saved result	0.000581921	0.01	Computed on submit

[Submit Analysis Time](#)

Live Analysis State

Use this table to track each hypothesis, the latest submitted result, and the next analysis time.

Hypothesis	Current Alpha	Last Submitted Analysis Time	Decision	In Graph	Testable	Next Analysis Time
H1	0.01		Pending	Yes	Yes	1
H2	0.015		Pending	Yes	Yes	1
H3	0		Not testable	Yes	No	2

[Submitted Analyses](#)

Submit interim p-values

Analysis Time 1:
interim p-values
before submit

Submit analyses
time when done

Case B, Analysis Time 1: Result

Choose an analysis time, then enter one-sided p-values for every active hypothesis scheduled at that analysis time.

Analysis Time

Analysis Time 2

Hypothesis	Interim Analysis (Look)	Current Alpha	Planned Information Fraction	Planned Max Info	Observed Info	Previous Saved Result	Boundary p	Entered One-Sided p	Decision
H1	Final Analysis 2 of 2	0.01	1	100	100	Analysis Time 1 / Look 1: p 0.01 vs 0.000269717	0.009907576	0.0001	Computed on submit
H2	Final Analysis 2 of 2	0.015	1	100					Computed on submit

Only active, testable hypotheses are shown here. Inactive rows are hidden.

Submit Analysis Time

Saved analysis time 1 with 2 saved results across 2 hypotheses. Now showing Analysis Time 2.

Result from the previous analyses

Live Analysis State

Use this table to track each hypothesis, the latest submitted result, and the next scheduled analysis time.

Hypothesis	Current Alpha	Last Submitted Analysis Time / Look	Last Observed p	Decision	In Graph	Testable	Next Analysis Time
H1	0.01	Analysis Time 1 / Look 1	0.01	Do not reject	Yes	Yes	2
H2	0.015	Analysis Time 1 / Look 1	0.01	Do not reject	Yes	Yes	2
H3	0			Not testable	Yes	No	2

Submitted Analyses

Case B Click-Through: Enter P-Values

Choose an analysis time, then enter one-sided p-values for every active hypothesis scheduled at that analysis time.

Analysis Time

Analysis Time 2									
Hypothesis	Interim Analysis (Look)	Current Alpha	Planned Information Fraction	Planned Max Info	Observed Info	Previous Saved Result	Boundary	Entered One-Sided p	Decision
H1	Final Analysis 2 of 2	0.01	1			Analysis Time 1 / Look 1: p 0.01 vs 0.000269717 Do not reject	0.009907516	0.0001	Computed on submit
H2	Final Analysis 2 of 2	0.015	1			Analysis Time 1 / Look 1: p 0.01 vs 0.000269717 Do not reject	0.0148041	0.02	Computed on submit

Only active, testable hypotheses are shown here. Inactive rows are hidden.

Submit Analysis Time

Saved analysis time 1 with 2 saved results across 2 hypotheses. Now showing Analysis Time 2.

Live Analysis State

Use this table to track each hypothesis, the latest submitted result, and the next scheduled analysis time.

Hypothesis	Current Alpha	Last Submitted Analysis Time / Look	Last Observed p	Decision	In Graph	Testable	Next Analysis Time
H1	0.01	Analysis Time 1 / Look 1	0.01	Do not reject	Yes	Yes	2
H2	0.015	Analysis Time 1 / Look 1	0.01	Do not reject	Yes	Yes	2
H3	0			Not testable	Yes	No	2

Submitted Analyses

Analysis Time 2: final p-values before submit

Case B Round 2 Analysis State

Graphical Multiple Testing

Home

Design

Group Sequential Design

Analysis

Locked design

Hypotheses: H1, H2, H3. Analysis times: 2. Design signature applied.

Analysis

Submit one full analysis time at a time. Enter one-sided p-values for every hypothesis scheduled at that analysis time and the app will update the graph and alpha recycling together.

Analysis Time = shared trial checkpoint. Interim Analysis (Look) = hypothesis-specific test at that checkpoint.

Open live graph and activity

Reset Analysis State

Submit One Analysis Time

Choose an analysis time, then enter one-sided p-values for every active hypothesis scheduled at that analysis time.

Analysis Time

Analysis Time 2

Hypothesis	Interim Analysis (Look)	Current Alpha	Planned	Decision
H3	Final Analysis 1 of 1	0.025	1	Computed on submit

Submit Analysis Time

Saved analysis time 2 with 2 saved results across 2 hypotheses.

Live Analysis State

Use this table to track each hypothesis, the latest submitted result, and the next scheduled analysis time.

Hypothesis	Current Alpha	Last Submitted Analysis Time / Look	Last Observed p	Decision	In Graph	Testable	Next Analysis Time
H1	0	Analysis Time 2 / Look 2	0.0001	Reject	No	No	—
H2	0	Analysis Time 2 / Look 2	0.02	Reject	No	No	—
H3	0.025			Pending	Yes	Yes	2

Submitted Analyses

Submission is the saved batch. Analysis Time / Look identifies the specific checkpoint and hypothesis-specific look inside that batch. One hypothesis can appear more than once in the same submission when same-analysis alpha recycling triggers a retest. Alpha At Submission is frozen for each saved event row and does not change after recycling.

Submission	Analysis Time / Look	Hypothesis	Alpha Spending	Info Fraction	Alpha At Submission	P	Boundary p	Boundary z	Decision
Submission 1	Analysis Time 1 / Look 1	H1	Lan-DeMets O'Brien-Fleming	0.5	0.01	0.01	0.000269717	3.46037	Do not reject
Submission 1	Analysis Time 1 / Look 1	H2	Lan-DeMets O'Brien-Fleming	0.5	0.015	0.01	0.0005819217	3.247596	Do not reject
Submission 2	Analysis Time 2 / Look 2 (final)	H1	Lan-DeMets O'Brien-Fleming	1	0.01	0.0001	0.0099070116	2.329851	Reject
Submission 2	Analysis Time 2 / Look 2 (final)	H2	Lan-DeMets O'Brien-Fleming	1	0.0225	0.02	0.0223499396	2.007468	Reject

Calculated and gave decision based on rejection boundaries

History of the submitted analyses

References and Acknowledgments

References

Bretz et al. (2009).

A graphical approach to sequentially rejective multiple test procedures.

Bretz et al. (2011).

Test and power considerations for multiple endpoint analyses using sequentially rejective graphical procedures.

Maurer and Bretz (2013).

Multiple Testing in Group Sequential Trials Using Graphical Approaches.

@Manual{,

title = {TrialSimulator: Clinical Trial Simulator},

author = {Han Zhang},

year = {2026},

note = {R package version 1.18.0},

url = {https://zhangh12.github.io/TrialSimulator/},

}

Team and support

- Phan Nguyen Huong Le, MengYang Yi
- Han Zhang, Philip He
- IDSWG Oncology Working Group
- Daiichi Sankyo, Inc.
- Rutgers University and Johns Hopkins University

Resources and Questions

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Scan



Scan for the app
or open the
public deployment

Thank you<3